

Exploratory Data Analysis Report on Hotel Booking Data



Business Problem

In recent years, both City Hotel and Resort Hotel have experienced elevated cancellation rates, leading to several challenges, including reduced revenue and suboptimal room occupancy.

As a result, minimizing cancellation rates has become the primary objective for both hotels to enhance their revenue-generating efficiency. This report aims to provide comprehensive business recommendations to address this issue.

The focus of this analysis will be on hotel booking cancellations and other factors that do not impact their business operations and annual revenue generation.

Assumptions

1. No significant anomalies occurred between 2015 and 2017 that would materially affect the data used in this analysis.
2. The information remains relevant and can effectively inform the hotels' strategic planning.
3. There are no unforeseen drawbacks associated with the implementation of any recommended strategies.

4. The hotels are not currently utilizing any of the proposed solutions.
5. Booking cancellations are the primary factor influencing revenue generation.
6. Cancellations lead to vacant rooms for the duration of the original booking.
7. Customers typically cancel reservations within the same year they were made.

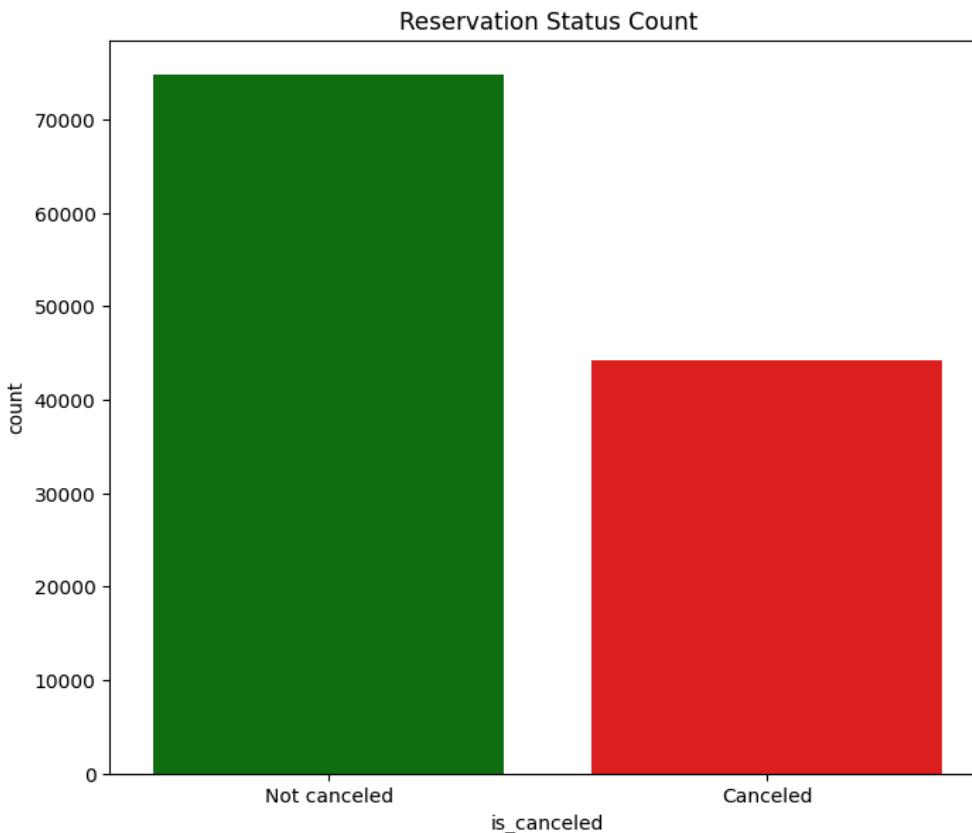
Research Questions

1. What factors influence hotel reservation cancellations?
2. How can we improve the management of hotel reservation cancellations?
3. In what ways can hotels be supported in making pricing and promotional decisions?

Hypotheses

1. Higher prices lead to an increase in cancellation rates.
2. Longer waiting lists are associated with more frequent customer cancellations.
3. Most clients are making their reservations through offline travel agents.

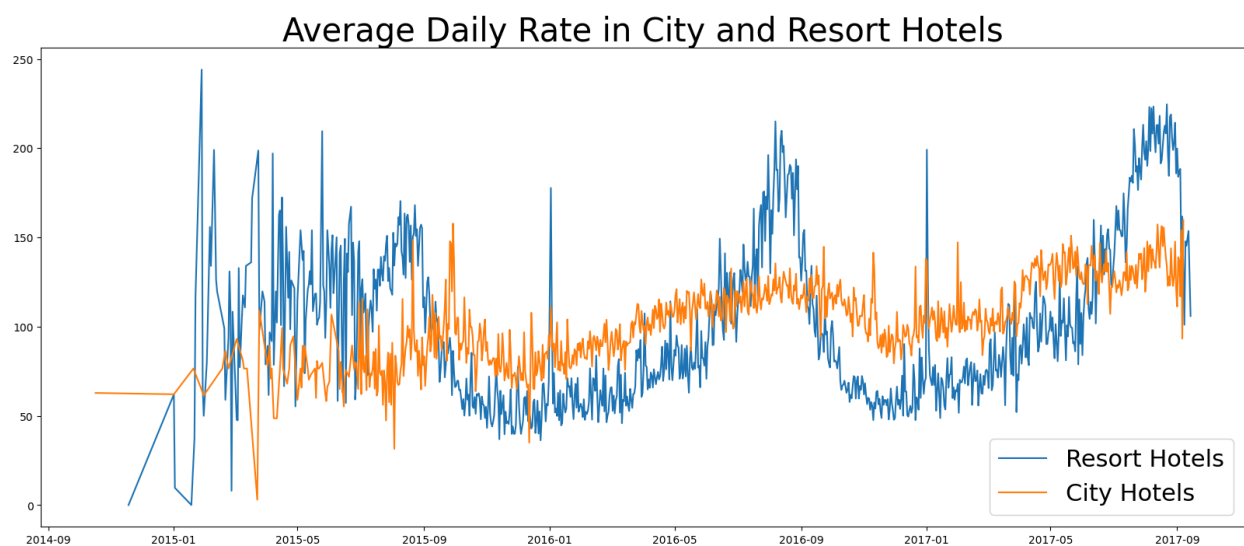
Analysis And Findings



- The accompanying bar graph illustrates the percentage of reservations that have been canceled compared to those that remain intact.
- It is evident that a considerable portion of reservations has not been canceled, with 37% of clients opting to cancel their bookings. This cancellation rate has a notable impact on the hotels' revenue.

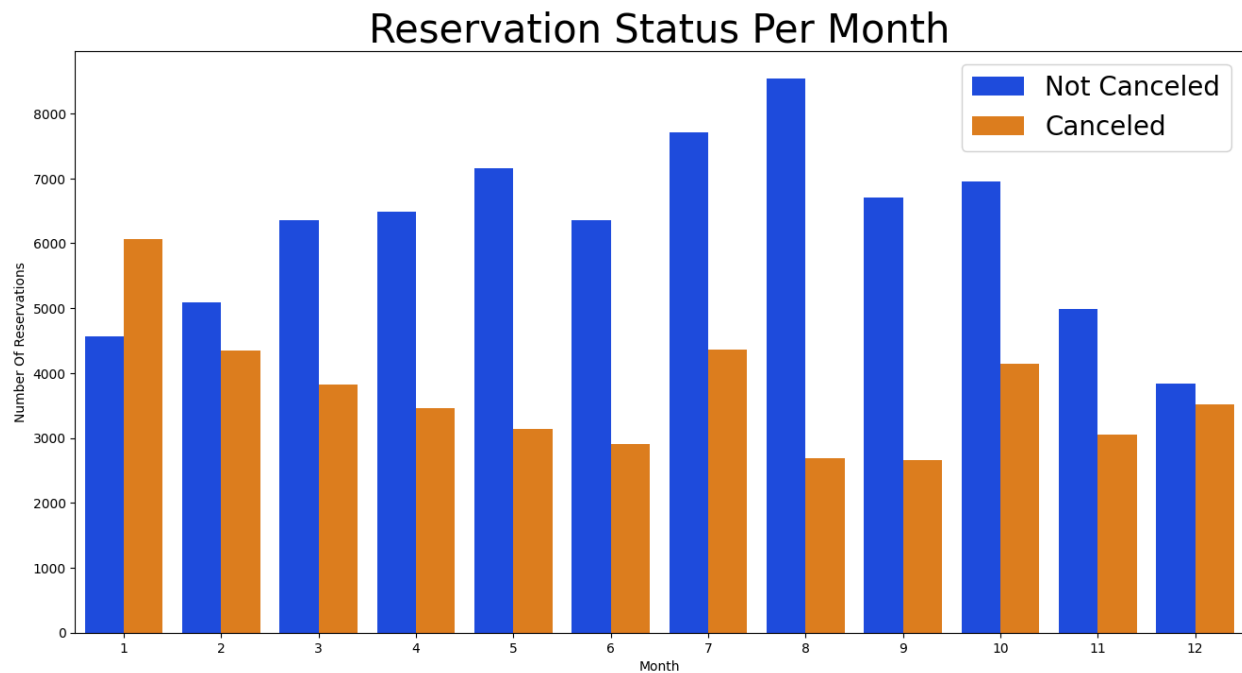


City hotels tend to have higher booking rates compared to resort hotels. This difference may be attributed to the higher prices often associated with resort hotels.



The line graph above indicates that, on certain days, the average daily rate for city hotels falls below that of resort hotels, and on other days, it is even lower.

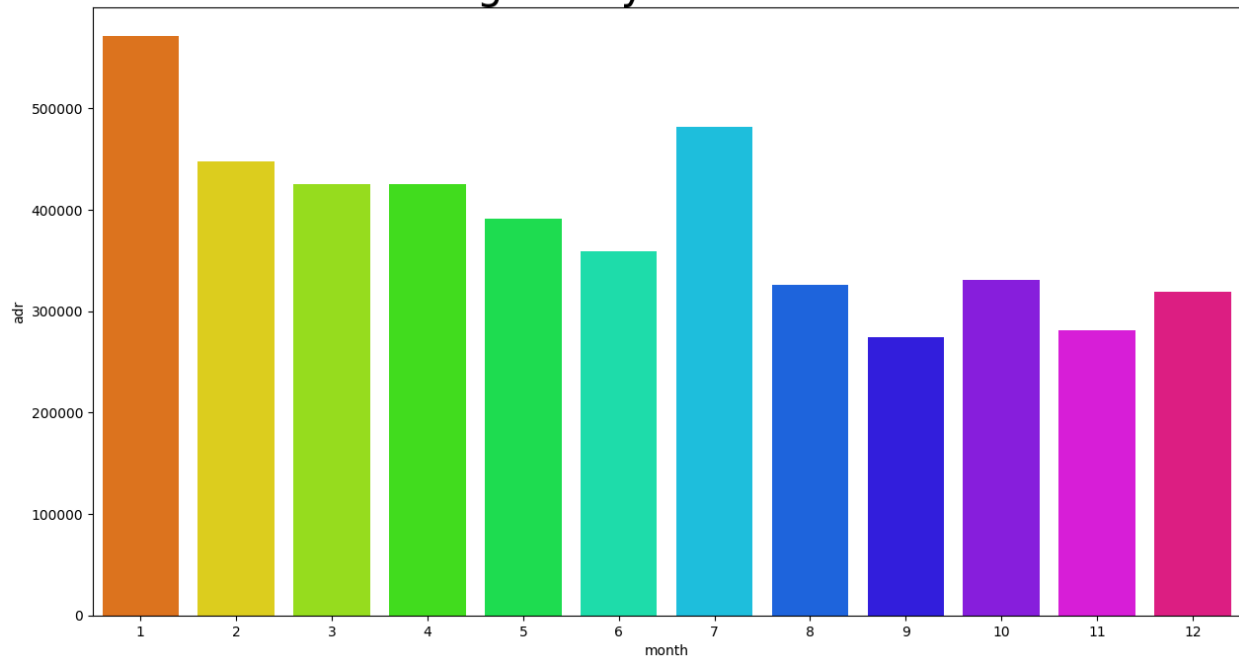
It is clear that rates for resort hotels may increase during weekends and holidays.



We created the grouped bar graph to examine the months with the highest and lowest reservation levels based on reservation status.

As shown, both confirmed and canceled reservations peak in September, while January records the highest number of cancellations.

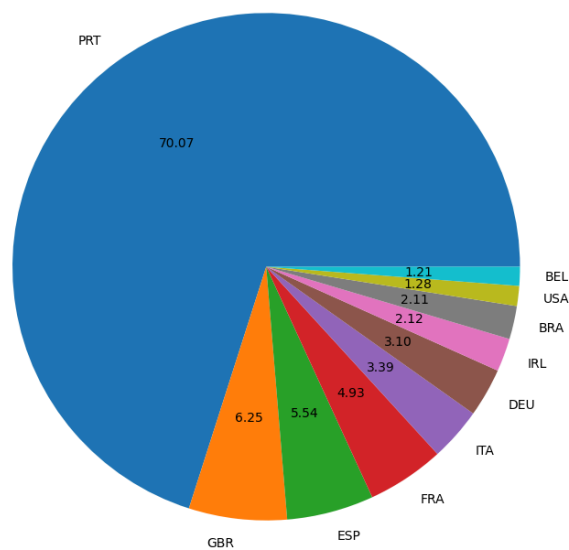
Average Daily Rate Per Month



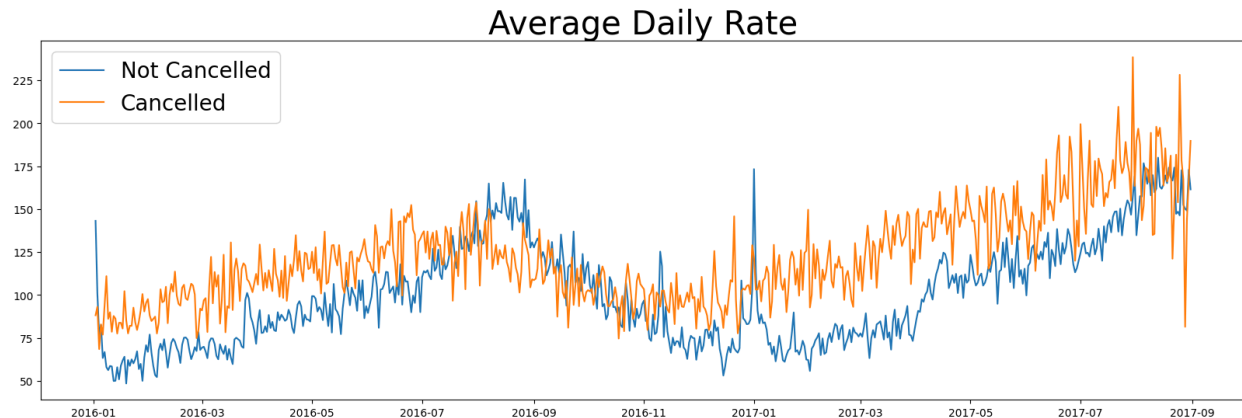
This bar graph illustrates that cancellations are most frequent when prices are highest and least frequent when they are lowest. Thus, the cost of accommodation is a key factor influencing cancellations.

Next, let's identify the country with the highest cancellation rate. Portugal leads with the highest number of cancellations.

Top 10 Countries with High Reservation Cancellation Rate



Let's examine the sources from which guests are making reservations at the hotels. Are they coming from direct bookings, groups, or online and offline travel agents? Approximately 46% of clients book through online travel agencies, while 27% come from groups. Only 4% of clients make reservations directly by visiting the hotels.



As shown in the graph, reservations are more likely to be canceled when the average daily rate is higher compared to when it is lower. This clearly supports the previous analysis that higher prices result in increased cancellations.

Suggestions

1. Cancellation rates increase as prices rise. To minimize reservation cancellations, hotels could adjust their pricing strategies by lowering rates for specific locations and offering discounts to consumers.
2. Since the cancellation rate is higher for resort hotels compared to city hotels, these hotels should consider providing reasonable discounts on room rates during weekends and holidays.
3. In January, when cancellations peak, hotels can launch targeted campaigns or promotions to boost revenue.
4. Additionally, improving the quality of accommodations and services, particularly in Portugal, can help reduce cancellation rates.